+1.8v, and as a result, will only return values up to their maximum operating voltages. The Arduino interface board, however, has a convenient pin called AREF located near digital pin 13 along with a function called `analogReference()` to give the Arduino's ADC a reference voltage other than +5v. This function will effectively increase the resolution available to analog inputs that operate at some other range of lower voltages below +5v. The syntax for this function follows.

```
analogReference(type)
```

The function is only called once in a sketch, but must be declared before `analogRead()` is used for the first time, making it a suitable candidate for placing in the `setup()` function. The type specified relates to the kind of reference voltage that we want to use. There are three possible reference voltage types on the Arduino Uno to be used for `analogReference()`, as listed in Table 6-1.

Table 6-1. *Analog Reference Types*

Type	Description
`INTERNAL`	Selects an internal reference voltage of +1.1v
`EXTERNAL`	Selects the external voltage reference connected to the AREF pin
`DEFAULT`	Returns to the default internal +5v voltage reference

While there are many reasons and methods for using a different voltage reference, many of these get very complex very quickly. The one we are most interested in providing is an external +3.3v reference when we use an accelerometer sensor to detect movement, discussed in greater detail in a later chapter. In order to get accurate readings from this sensor, we will make the following statement once at the beginning of our source code:

```
analogReference(EXTERNAL);
```

We will then need to connect the AREF pin to the output pin labeled 3.3V using a jumper wire. This will use the Arduino's Uno secondary onboard voltage regulator to provide an accurate external analog reference of +3.3v. With this setup, it is important to call the `analogReference()` function before the `analogRead()` function, or it is possible to short the internal and external voltages—perhaps damaging the microcontroller. Finally, it is also important to remember to never connect the AREF pin to a voltage source greater than +5v because it could also damage the microcontroller.

Analog Serial Monitor

So far, we have made the general assumption that the readings we are receiving from the analog input pins are providing a range of values from 0 to1023, but this is rarely the case, as you can see in our opening project's source code. The reason for this is that every sensor is a little different and will provide a varying range of voltages depending on what they are sensing. In order to better understand the range of values that a particular sensor is providing, we need to create a sketch to give us some visual representation of the values reported by the ADC. Before getting to that, let's back up again and look at a simpler analog sensor, as shown in Figures 6-5 and 6-6.